MGMT510 - Total Quality Management (original code=MGT510) Faqs For Final Term Exam Preparation

FAQs

Question: What is the significance of TQM?
Answer: Significance of TQM The importance of TQM lies in the fact that it encourages innovation, makes the organization adaptable to change, motivates people for better quality, and integrates the business arising out of a common purpose and all these provide the organization with a valuable and distinctive competitive edge. Through TQM, companies will be ready to achieve values such as: • High product/service value/quality • Satisfactory product/service in a long life-cycle • Efficiency in lead-time and cycle time • High competitive value • Economical and quick response to emergencies

Question: How cost can be reduced by improving quality?
Answer: Cost Reduction via Quality Improvement: Generally the perception is that quality improvements may result in increase in cost. Our strong belief is that the implementation of quality drive will certainly result in achieving cost reduction. In this respect some aspects of cost of quality are given below: 1. Cost of Conformance a) Cost of Prevention b) Cost of Appraisal 2. Cost of Non-Conformance or Failures a) Cost of Internal Failure b) Cost of External Failure or lost opportunity The following explanations will elucidate our earlier point that cost reduction will follow if quality improvements are ensured. 1(a) Cost of Prevention It is the cost to ensure that customer requirements are met. Examples include: • Quality Planning • Quality Engineering • Training to improve quality • Maintenance & Calibration of production & inspection of equipment • Supplier assurance The objective is to ensure maintenance of quality systems. (b) Cost of Appraisal It is the cost to ensure that the work processes are producing outputs that meet customer needs or requirements. Examples include: • Quality data acquisition and analysis • Quality measurement criteria • Quality audits • Laboratory Acceptance Testing • Field Evaluation and testing • Inspection and Testing • Raw Materials Testing • In-process Testing • Review of test and inspection data The above expenses are incurred after production, but before sales to identify defective items.

Question: What is the difference between TQM and Strategic TQM?

Answer: TQM is the attainment of customer satisfaction through continual process improvement. Strategic TQM implies formulating plans to achieve the very goal that is, customer satisfaction.

Question: What is the difference between 'Big Q' and 'Little Q'?
Answer: 'Big Q' is a broader term that refers to the quality of products, services, people, processes and environments as a whole. It considers all these elements as a system. 'Little Q' is a narrower term referring to the quality of each of these elements individually.

Question: What is the significance of ISO 9000?
Answer: ISO 9000 is International organization for standardization 9000 means a set of quality management standards, which are completed in 1982 and published in 1983. Now more than 90 countries have adopted ISO 9000 2000 as national standard. ISO 9000 series consists of following three standards: - ISO 9000:2000 Quality Management systems – Fundamental - ISO 9001:2000 Quality Management systems – Requirements or specifications - ISO 9004:2000 Quality Management systems – Guidance for Performance Improvement The various benefits for adopting these standards are: • Quality of product or service improves • Management system improves • Less customer complaints indicates Customer satisfaction • Waste elimination • Internal communication of your organization improves • Market opportunity increases If your organization is already ISO certified then it's your responsibility that you should maintain quality and thus in turn maintain the standards. But if your organization is not ISO quality management system certified then you must ascertain your organization's plan so as to achieve ISO certification. When someone purchases a product from an organization which has ISO 9000 standard certification with appropriate standard then the customer has assurance that the quality he receives is what he is expecting.

Question: What is the difference between continuous and continual quality improvement?
Answer: Continual improvement is a broader term preferred by W. Edwards Deming to refer to general processes of improvement and encompassing "discontinuous" improvements—that is, many different approaches, covering different areas. Continuous improvement is a subset of continual improvement, with a more specific focus on linear, incremental improvement within an existing process. Some practitioners also associate continuous improvement more closely with techniques of statistical process control.

Description: PIQC Profile,newsletter and case studies

Description: Quality magazine and discuss about different concept of quality

MGMT510 - Total Quality Management (original code=MGT510) Glossary For Final Term Exam Preparation

Glossary

Aesthetics : Aesthetics is commonly known as the study of sensory or sensori-emotional values, sometimes called judgments of sentiment and taste. More broadly, scholars in the field define aesthetics as "critical reflection on art, culture and nature. It examines what makes something beautiful, sublime, disgusting, fun, cute, silly, entertaining, pretentious, discordant, harmonious, boring, humorous, or tragic.

Affinity Diagram : A way to organize idea data into coherent patterns or themes. A large number of ideas are generated and then organized into groupings to reveal major themes.

Authoritarian Culture : An organizational culture characterized by the holding of all power (decision making and information) at the top of the organization. The authoritarian organization seeks to maintain the status quo and forces workers to conform, never question or give feedback, play politics, and wait for orders.

Benchmarking : A systematic process of continuously measuring an organization's critical business processes against business leaders anywhere in the world to gain information which will help the organization take action to improve its performance. Steps include Planning the Study, Collecting Information, Analyzing Results, Implementing Improvements.

Boundary : The beginning or end point in the portion of a process from a Supplier to a Customer that will be the focus of the process improvement effort.

Brainstorming : A group decision-making technique designed to generate a large number of creative ideas through an interactive process. Brainstorming is used to generate alternative ideas to be considered in making decisions.

Center Line : The line on a control chart that represents the average (mean or median) value of the items being plotted.

Check Sheet : A data collection form consisting of multiple categories. Each category has an operational definition and can be checked off as it occurs. Properly designed, the Check Sheet helps to summarize the data, which is often displayed in a Pareto Chart.

Coach : A key resource person from within the organization who will support the CEO's leadership of the CQI. A respected peer from the hospital work force who is enthusiastic and knowledgeable about CQI, eager to learn and eager to help others learn.

Collaborative Culture : An organizational culture characterized by a shared vision, shared leadership, empowered workers, cooperation among organizational units as they work to improve processes, a high degree of openness to feedback and data, and optimization of the organizational whole versus its many parts.

Common Cause System of Variation : The collection of variables that produce common cause variation and the interaction of those variables.

Continuous Quality Improvement (CQI) : The culture, strategies and methods necessary for continual improvement in meeting and exceeding customers' expectations.

Control Chart : A display of data in the order that they occur with statistically determined upper and lower limits of expected common cause variation. It is used to indicate special causes of process variation, to monitor a process for maintenance, and to determine if process changes have had the desired effect. One of the basic tools of the New Quality Technology.

Control limits : Expected limits of common cause variation. Sometimes they are referred to as upper and lower control limits. They are not specification or tolerance limits.

Cross Functional Approach : According to old paradigm, centuries old hierarchical form of organization with a vertical chain of command was the norm until recently emergence of new paradigm), when organizational complexity demanded horizontal as well as vertical coordination in order to plan and control processes that flowed laterally. If no lateral coordination is achieved, the organization becomes a collection of islands of specialization without integration, a requirement of systems approach. There is widespread agreement that cross functional teams provide the best vehicle for linking organizational activities and processes.

Customer : The receiver of an output of a process, either internal or external to an organization or corporate unit. A customer could be a person, a department, a company, etc.

Customer Data Table : A tool for translating customers' words into requirements, quality indicators and features of the product or service.

Data Collection : Gathering facts on how a process works and/or how a process is working from the customer's point of view. All data collection is driven by knowledge of the process and guided by statistical principles.

Deming Cycle : Deming Cycle for Continuous Improvement is a visualization of the CQI process usually consisting of four points - Plan, Do, Check, Act -- linked by quarter circles. The cycle was first developed by Dr. Walter A. Shewhart but was popularized in Japan in the 1950 by Dr. W. Edwards Deming.

Deming's 14 Principles : The foundation of Deming's philosophy. The points are a blend of leadership, management theory, and statistical concepts which highlight the responsibilities of management while enhancing the capacities of employees.

European Foundation for Quality Management (EFQM) : EFQM stands for European Foundation for Quality Management Excellence Model. This framework was the first one to include 'Business Results' and to really represent the whole business model. Like the Baldridge, the EFQM model recognized that processes are the means by which an organization harnesses and releases the talents of its people to produce result/performance. Moreover, improvement in performance can be achieved only by improving the processes by involving the people.

Facilitator/Advisor : A person who has developed special expertise in the CQI process. In a CQI team, the facilitator/advisor is not a team member but a person outside the group who serves as a process guide, teacher of CQI methods, and consultant to the team leader, and who helps connect the work of the team to the organization's overall CQI effort.

Flowchart : A graphical representation of the flow of a process. A useful way to examine how various steps in a process relate to each other, to define the boundaries of the process, to identify customer/supplier relationships in a process, to verify or form the appropriate team, to create common understanding of the process flow, to determine the current "best method" of performing the process, and to identify redundancy, unnecessary complexity and inefficiency in a process.

FOCUS-PDCA - : A strategy that provides a roadmap for continuous process improvement when linked to a quality definition. It is an acronym meaning: Find a process to improve, Organize a team that knows the process, Clarify current knowledge of the process, Understand sources of process variation, Select the process improvement, Plan the improvement and continued data collection, Do the improvement, data collection, and analysis, Check and study the results, Act to hold the gain and to continue to improve the process.

Force Field Analysis : A systematic method of understanding competing forces that increase or decrease the likelihood of successfully implementing change.

Future State : In an organizational transformation, the vision of where the organization will be after it is transformed. For the transformation to CQI, the future state includes constancy of purpose, leaders who model the new way, collaboration, customer mindedness, and a process focus.

Immediate customer : The person or unit that directly receives the output of the process.

Input : The service or product a supplier provides to a process. Inputs to one process are the outputs from preceding processes.

Interrelationship Digraph : A way to display cause-and-effect relationships among all the elements in a system. The relationship arrows indicate the issues/causes that are the most fundamental among all the related items.

Ishikawa Diagram : A graphic tool used to explore and display all the factors that may influence or cause a given outcome. (Also known as a cause and effect or fishbone diagram.)

Key Process Variable : A component of the process that has a cause and effect relationship of sufficient magnitude with the Key Quality Characteristic such that manipulation and control of the KPV will reduce variation of the KQC and/or change its level.

Key Quality Characteristic : The most important quality characteristics. The KQCs must be operationally defined by combining knowledge of the customer with knowledge of the process. KQCs are measured to understand the actual performance of the process.

Median : In a series of numbers, the median is a number which has at least half the values greater than or equal to it and at least half of them less than or equal to it.

Meeting Process : A defined method for conducting meetings that includes specific roles and responsibilities for a team leader, a recorder, a timekeeper, team members, and a facilitator or advisor. The steps are 1) Clarify the objective, 2) Review roles, 3) Review the agenda, 4) Work through agenda items, 5) Review the meeting record, 6) Plan next steps and next meeting agenda and 7) Evaluate.

Mentor : A highly skilled CQI professional with extensive training and experience in the initiation and operation of CQI. A resource person from outside the organization or department who visits periodically to counsel the CEO, Coach and Quality Improvement Council in the development, implementation and evaluation of CQI.

Multiple Voting : A group decision-making technique designed to reduce a long list to a few ideas.

Nominal Group Brainstorming : A group process technique designed to efficiently generate a large number of ideas through input from individual group members.

Operational Definition : A description in quantifiable terms of what to measure and the steps to follow to measure it consistently. Deming has suggested that a good operational definition includes: 1) a criterion to be applied, 2) a way to determine whether the criterion is satisfied, and 3) a way to interpret the results of the test. An operational definition is developed for each KQC or process variable before data is collected.

Opportunity Statement : A concise description of a process in need of improvement, its boundaries, and the general area of concern where a CQI Team should begin its efforts.

Outcome (Benefit) : The degree to which Outputs meet the needs and expectations of the Customer.

Output : The service or product that a customer receives from a process. The output of one process can be the input to a succeeding process.

Owner : The person who has or is given the responsibility and authority to lead the continuing improvement of a process. Process ownership is a designation made by leaders of organizations and depends on the boundaries of the process.

Paradigm Shift : A point in time when the knowledge or structure which underlies a science or discipline changes in such a fundamental way that the beliefs and behavior of the people involved in the science or discipline are changed.

Pareto Chart : A bar graph used to arrange information in such a way that priorities for process improvement can be established. It displays the relative importance of data and is used to direct efforts to the biggest improvement opportunity by highlighting the vital few in contrast to the many others.

Penny Matrix : A way to prioritize a list of options by pooling the opinions of raters. Raters "spend" their pennies across several options with the sums of "money spent" indicating a priority weighting and ranking to the options.

Perceived Quality : It can be defined as the customer's perception of the overall quality or superiority of a product or service with respect to its intended purpose, relative to alternatives. Perceived quality is, first, a perception by customers.

Present State : In a force field analysis, the description of an organization as it currently exists. It includes what happens in the organization, both formally and informally.

Prioritization Matrix : A way to prioritize options by requiring the raters to work to consensus on priorities. Also known as a paired-comparison technique., the tool pairs each option with each other, with row totals indicating weighting and ranking.

Process : A series of actions which repeatedly come together to transform Inputs provided by a Supplier into Outputs received by a Customer. A process can be used to develop products and services.

Process Decision Program Chart (PDPC) : A tool for improving implementation through contingency planning. By considering "what could go wrong?", plans for prevention of problems are generated.

Process Improvement : The continuous endeavor to learn about all aspects of a process and to use this knowledge to change the process to reduce variation and complexity and to improve customer judgments of quality. CFI begins by understanding how customers judge quality, how processes work, and how understanding the variation in those processes can lead to wise management action.

Process Variation : The spread of process output over time. There is variation in every process, and all variation is caused. The causes are of two types - special or common. A process can have both types of variation at the same time or only common cause variation. The management action necessary to improve the process is very different in each situation.

Quality Assurance (QA) : The activities include a planned system of review procedures conducted by personnel not directly involved in the inventory compilation/development process.

Quality Characteristics : Characteristics of the output of a process that are important to the customer. The identification of quality characteristics requires knowledge of the customer needs and expectations.

Quality Control (QC) : It is a system of routine technical activities, to measure and control the quality of the inventory as it is being developed.

Quality Improvement Council (QIC) : A group composed of the Coach and the senior leadership of an organization which is primarily responsible for planning, strategy development, deployment, monitoring, educating, and promoting CQI.

Quality Inspection : Usually consists of three stages - sampling, measuring, and sorting. While many organizations rely on inspection to improve quality, the better way is to design quality into the product or service - to improve the process. This may include some inspection as a means of data gathering.

Quality Planning/Redesign : Creating new or redesigned products/services/processes to meet customer requirements. The steps of this method are Organize the project, Identify key customers, Determine requirements, Establish quality indicators, Design, Strengthen the design, Test the design, Implement and improve.

Red Bead Experiment : A simple exercise to demonstrate, among other things, that many managers hold workers to standards beyond their control, variation is part of any process, and workers work within a system beyond their control. The game also shows that some workers will always be above average, some average, and some below average, that the system, not the skills of individual workers, determines to a large extent how workers in repeating processes perform, and that only management can change the system or empower others to change it.

Refreezing : Recognizing, reinforcing, and rewarding new organizational attitudes and behaviors so they become the norm. Making processes, systems, and methods throughout the organization support CQI.

Requirement-Indicator Matrix : A matrix that shows the presence of all possible relationships between customer requirements and quality indicators.

Rework : The act of doing something again because it was not done right the first time. It can occur for a variety of reasons, including insufficient planning, failure of a customer to specify the needed input, and failure of a supplier to provide a consistently high quality output.

Role Definition : Members of a successful organization need a sound understanding of their roles during transition to a TQM program. People at all levels require orientation as to how they will be impacted under the new philosophy of employee involvement. The improvement process involves a group of complementary activities that provide an environment conducive to improvement of performance for both employees and mangers. Each level has role to play.

Run : A point or a consecutive number of points that are above or below the central line in a run chart. Too long a run or too many or too few runs can be evidence of the existence of special causes of variation.

Run Chart : A display of data in the order that they occur. Run charts display process variation and can be used to indicate special causes of process variation in the form of trends, shifts, or other non-random patterns.

Special and Common Cause System of Variation : The collection of variables that produce both common cause variation and special cause variation and the interaction of those variables.

Special Cause Variation : Variation in the process that is assignable to a specific cause or causes. It arises because of special circumstances.

Spider Diagram : A visual report card for the performance of a number of indicators on a single chart. Also know as a "radar chart" and a "gap analysis" tool, this diagram makes visible the gaps between the current and desired performance.

Status quo : Status quo means “current situation”. A preference for the status quo means a reluctance to change.

Supplier : The party or entity responsible for an input to a process. A supplier could be a person, a department, a company, a nursing school, etc.

Tampering : Taking action without taking into account the difference between special and common cause variation.

Team Leader : A person designated to lead the CQI Team. An individual who has team leadership skills and basic quality improvement skills.

Teams : • Cross-functional - A group of usually five to eight people from two or more areas of the organization who are addressing an issue which impacts the operations of each area. For example, the processes of meeting information requests might be addressed by a team involving PI,, managed care and marketing staff. • Functional - A group of five to eight people addressing an issue where any recommended changes would not be likely to affect people outside the specific area. For example, a Functional Team concerned with filing and retrieving data in the laboratory might consist just of people who work in the lab.

Transformation : A major organizational change from the present state to a new/preferred state in which CQI flourishes. The primary steps involved in moving an organization through a transformation are present state, unfreezing, transition period, refreezing, and new/preferred state.

Transition Period : A description of the time when an organization is visibly moving away from the old way toward the new way. During this time, employee attitudes and behaviors range from being excited and busy to being confused and resistant. The support for change is building. New leaders emerge, champions of the change come forward and confusion over roles begins to clear.

Tree Diagram : A tool to expand a proposed change from a general idea to a specific series of concepts or actions. Used to systematically map out in increasing detail the full range of paths and tasks that need to be accomplished to achieve a primary goal and related subgoals.

Ultimate Customer : The person or unit who receives the output from a series of processes and for who these processes are designed. Without the ultimate customer, there would be no need for the intermediate processes to exist.

Unfreezing : Reassessing old values and behaviors and becoming open to the acceptance of a new culture.

MGT510 - Total Quality Management Finalterm Solved Subjective Paper July-2013

Why Statistical Process Control is carried out. Also describe the two methods for the improvement of yield of a process (i.e., taking it to a higher sigma value). (Marks=2+1)

Solution:

Statistical process control is carried out to record, identify and control any nonconformance of the product or service to the customer. Statistical process control tools help to record and control by identifying the level of non conformance. These tools help us to see the acceptable level of customer requirements and the actual output of the process.

Statistical process control tools only identify the non conformance. The improvement in the process is done by taking the corrective actions. After identifying that the output of a system is non conforming, specific corrective action are taken to adjust the process, then the new records will identify that either the process is under control or not. In this way the yield of a process can be enhanced.

Define a "team". Enlist some of the qualities an individual must possess as a team member. (Marks=1+2)

Solution:

Team is a collection of 5 to 7 individuals having interdependent and inter related responsibilities and are combined to perform a specific task.

Individual must have;

Shared values

Good communication skills

Desire for ultimate success

Team work spirit

Understanding of synergies produced in team work

Self confidence

Self motivation

Ability to follow as well as lead

What are the responsibilities of the owner of a critical success factor?

Solution:

The owner of critical success factor is responsible to make the success of the factor in the best way possible with in the specific time and in the minimum cost.

The owner of a critical success factor is responsible to understand the importance of overall system and its relation to the success factor he owns.

He must be aware of the importance of inter-relationship among the organizational activities and the reliance of one activity on others.

He must be responsible to be committed and ensure the commitment of individual working with him, towards the over all success of the organization.

Justify the statement, "All strategic decisions made by a company are customer-driven"?

Solution:

The statement can be justified by understanding the step which an organization takes while designing the steps that it decides to take for doing the business.

First of all each business do the market research to identify the feasibility of the business and to know the potential customer, then do research on the needs and wants of the customers, their likes dislikes, norms values profession all the relevant factors are studied, then strategic decision are taken.
So we can say that in today's world none of the business can survive by ignoring the customer preferences. The customer is treated as the king of over all organizational process. Customer focused organization made each and every decision by keeping their customers in mind, because they think that what ever they do is just to satisfy their customers.
So in conclusion we can say that it is not the top management which drives the organizations, actually top managers identify the customer requirements and expectations and then drive the organization according to those requirements so that customer can be satisfied.

TQM is an organization-wide effort. How would you illustrate the role of middle management as far as the implementation of a Quality strategy is concerned?
Solution:
The TQM is an organization wide approach in which each and every member of the organization is supposed to understand the importance and benefits of the quality of process and products. The role of middle management in this regard is just like a bridge in the organization's conceptual and technical staff. The top management because of their long vision and high conceptual skills formulate the overall mission and goals of the organization. These quality goals are translated in to short term objectives by middle management and then communicated to the operational level. The middle managers have both the technical and conceptual skills with good communication skills, due to which they are able to understand and communicate at both levels. In this way they act like a bridge. They report the operational performance in term of goals achievement to the top management and also translate the top management goals in specific short term goals to the operational level and encourage their commitment for quality behavior and reveal the importance and benefits of the quality in the language they can understand. In this way the commitment is ensured from top to bottom.
What can be the possible sources of variation in production process? Give any five examples.
Source of variation can be attributed to two types of causes.
Common causes
Special causes
Common causes of variation:
These are the causes which are inherent in the system and the variations produced by these causes remain within the acceptable level of customer requirements. These causes are not assignable to any specific event.
Example:
All the eggs in the poultry farm lay one egg rotten in a week
The delivery of pizza house is always 5 minutes late.
The reading of a machine varies with a limit because of voltage fluctuations.

The air pressure varies during experiment in open area.
Special cause of variation:
These are the causes which are not inherent in the system and the variation produced by these causes crosses the acceptable level of customer's requirements. These causes can be assigned to an identifiable factor or source of the cause.
Example:
Only one egg laying rotten eggs
The delivery of a pizza is late for the first time.
The scientist is not taking the reading from proper angle and the reading crosses acceptable limits.
The instrument having a fault during the experiment due to which the figure arises in unexpected.

Being CEO of ABC Company, what are some of the pre-requisites you should ensure for successful empowerment in your company?

Solution:
Before empowering employee following things must be ensured.
The understanding of overall system.
The mental capability of employee
The sense of responsibility the employee has.
The level of trust the employee has on himself.
The level of confidence and commitment.
The level of confidence and commitment.
The ethical behavior of the employee.
the commitment of employee for the success of organization.
The provision of necessary training regarding the authority provided to the employee.

What is the significance of quality to gain competitive advantage?

Solution:
Quality is one of the strong and appealing factors which help the organization to gain the competitive edge over the competitor. The highest the level of quality the greater will be the customer satisfaction, which will ultimately result in customer loyalty and higher market share. In this ways the business will excel and the organization will prosper. The organization try to improve quality and work for quality standards certification, because it provide wide customer acceptance and open gates of global market for the organization.
So we can say that the significance of quality is like the most desirable feature of the organization because in this competitive era, the success is inherent in quality not in the quantity.